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CHAPTER 1
INTRODUCTION

1 INTRODUCTION

1.1. MARITIME GEOGRAPHY

The geopolitical architecture of South Asia is fundamentally dictated by the maritime geography of the Indian Peninsula. Extending deep into the Indian Ocean Region (IOR), the Indian landmass acts as a massive natural pier splitting the northern expanse of the ocean into two distinct hydrographic bodies: the Arabian Sea to the west and the Bay of Bengal to the east. This geographic configuration endows India with a highly vulnerable coastline of 7,516.6 kilometers, encompassing nine coastal states, two union territories, and two major island archipelagos—the Lakshadweep Islands in the west and the strategic Andaman and Nicobar chain in the east.

From an economic perspective, India relies on the maritime domain for national survival. Over 95% of the nation's trade volume and more than 68% of its total value are transported via merchant shipping lanes. More critically, India imports over 80% of its crude oil requirements, the vast majority of which passes through deep-sea lanes before arriving at critical domestic ports such as Mumbai, Mundra, Paradip, and Visakhapatnam. The United Nations Convention on the Law of the Sea (UNCLOS) grants India an Exclusive Economic Zone (EEZ) that covers over 2.02 million square kilometers. This vast area contains rich fisheries, extensive polymetallic nodules, and vital offshore energy infrastructure, including the Mumbai High and Krishna-Godavari basin oil fields.



Figure 1.1Indian Ocean Region and Key Maritime Links

Beyond domestic concerns, India's strategic position straddles the world's primary maritime Sea Lines of Communication (SLOCs). The global maritime economy depends on three critical choke points bordering the IOR: the Strait of Hormuz (facilitating the transit of Middle Eastern hydrocarbons), the Bab-el-Mandeb (connecting the Red Sea to the Mediterranean via the Suez Canal), and the Strait of Malacca (the gateway to the South China Sea and the Pacific Ocean). Approximately 70% of global oil trade and 50% of the world's container traffic traverse these waters annually. Consequently, any disruption to these lines—whether from state-sponsored blockades, asymmetric piracy, or

localized conflict—poses a direct threat to India's economic stability and regional energy security.

1.2. FLEET CAPABILITIES AND GAPS

The Indian Navy operates to project power, ensure sea control, and maintain Maritime Domain Awareness (MDA) across the IOR.

1.2.1. Core Strengths

(a) **Carrier Aviation and Blue-Water Projection:** The Indian Navy is one of a select few global forces operating multiple active Carrier Strike Groups (CSGs). The operational deployment of INS *Vikramaditya* and the indigenous INS *Vikrant* gives the navy a distinct advantage in projecting organic air power, enforcing wide-area sea control, and providing comprehensive air defense shields far beyond the range of land-based aircraft.



Figure 1.2 INS Vikrant aircraft carrier at sea

(b) **Stealth Platform Development:** Indian design capabilities have progressed through projects like the *Kolkata* and *Visakhapatnam*-class (Project 15A/B) guided-missile destroyers and the *Shivalik* and *Nilgiri*-class (Project 17/17A) stealth frigates. These ships show advanced mastery of radar cross-section (RCS) reduction techniques, infrared signature suppression, and network-centric warfare through integrated Combat Management Systems (CMS).

(c) **Material and Technological Infrastructure:** Collaborations between the Defence Research and Development Organisation (DRDO) and the Steel Authority of India Limited (SAIL) led to the domestic production of DMR 249A

and DMR 249B high-tensile warship-grade steel. This development eliminated reliance on foreign suppliers for structural hull plates.



Figure 1.3 INS Imphal Project 15B destroyer

(d) **Fiscal and Operational Inter-operability:** Decades of executing highly integrated bilateral and multilateral exercises (such as Malabar and Milan) have optimized tactical communication links and cross-deck helicopter logistics, positioning the Indian Navy as a primary security provider in the region.

CORE STRENGTHS		STRUCTURAL VULNERABILITIES	
1	Carrier strike group capability	1	Fleet depletion in littoral hulls
2	Advanced domestic stealth tech	2	Critical submarine shortages
3	Indigenous DMR 249A steel production	3	High operational asset fatigue
4	High inter-operability standards	4	Deficit in mine counter-measures

1.2.2. Structural Vulnerabilities

(a) **Asset Distribution Overstretch:** A critical gap exists between heavy blue-water combatants (destroyers and frigates) and light coastal assets. Using a 7,000-tonne stealth destroyer for routine anti-piracy operations, maritime border enforcement, or escort duties causes unnecessary wear and tear on complex machinery and expends disproportionate amounts of fuel.

(b) **Under-sea Warfare Shortages:** The conventional diesel-electric submarine fleet faces block obsolescence. Delays in procurement

frameworks have caused a dip in the minimum required hull count, limiting the navy's ability to maintain continuous underwater surveillance at key maritime choke points.

(c) **Mine Counter-Measures (MCM) Shortfall:** The fleet has a noticeable deficit in dedicated Mine Counter-Measure Vessels (MCMVs). This lack of modern mine-hunting sonar systems and autonomous underwater vehicles (AUVs) leaves vital commercial ports and naval bases vulnerable to asymmetric naval mining.

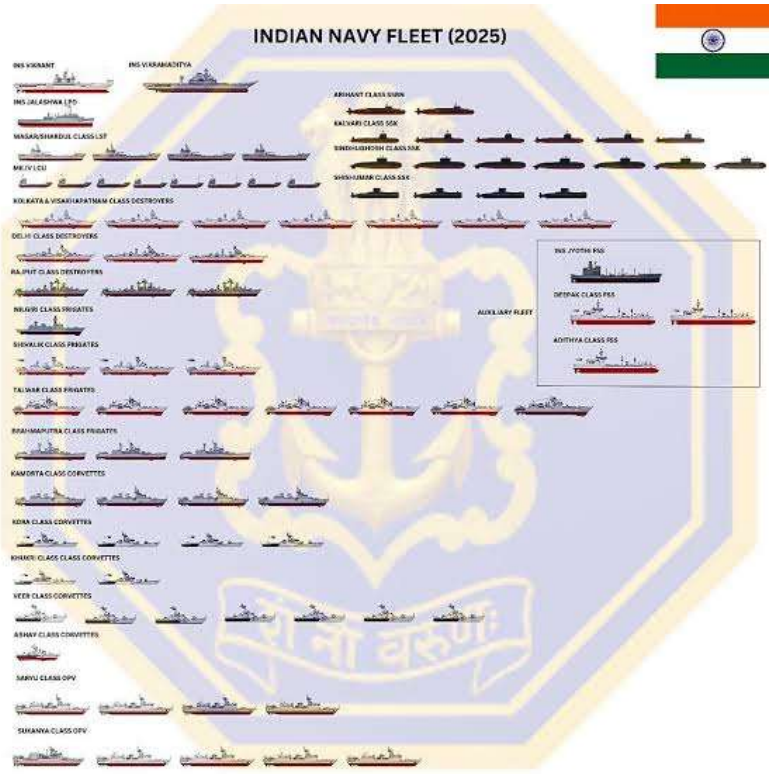


Figure 1.4 Indian Navy active fleet composition chart

(d) **Littoral Defense Gaps:** Legacy missile boats (such as the *Veer*-class) and older corvettes (like the *Kora* and *Khukri*-class) lack advanced anti-submarine sonar suites, vertical launch surface-to-air missile systems, and modern stealth geometries. This leaves them vulnerable in highly contested, modern anti-access/area-denial (A2/AD) environments.

1.3. THREAT MATRIX AND SECURITY CHALLENGES

India’s maritime operating environment is shaped by both traditional state rivalries and evolving asymmetric threats. The northern Indian Ocean is increasingly contested, requiring a highly adaptable surface fleet.

1.3.1. Traditional State Threats

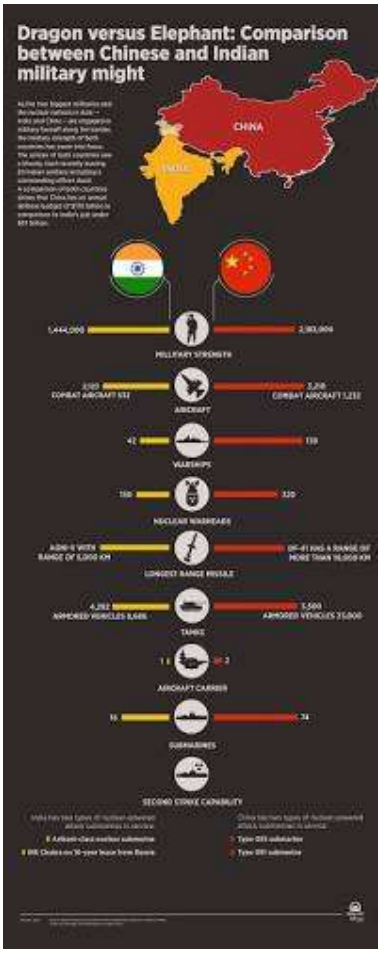


Figure 1.5 India-China military and naval strength comparison infographic

(a) **People's Liberation Army Navy (PLAN) Expansion:** The Chinese navy has expanded its presence in the IOR through its "String of Pearls" strategy. This involves establishing dual-use commercial and military port facilities in Gwadar (Pakistan), Hambantota (Sri Lanka), and Djibouti on the Horn of Africa. PLAN deployments include nuclear-powered attack submarines (SSNs), conventional hunter-killer submarines (SSKs), satellite-tracking ships, and hydrographic research vessels. These assets operate inside India's maritime zone to map thermal layers that affect acoustic detection.

(b) **Pakistan Navy Force Modernization:** Operating along India's western flank, the Pakistan Navy has focused on building a specialized underwater denial force. The acquisition of advanced Hangor-class conventional submarines equipped with Air-Independent Propulsion (AIP) allows these vessels to remain submerged for extended periods. This presents a major challenge to surface forces in the shallow, noisy hydroacoustic environments of the ArabianSea.



Figure 1.6 China and Pakistan 'Sea Guardians' bilateral naval exercise

1.3.1. Asymmetric Security Threats

(a) **Anti-Ship Cruise Missile (ASCM) Proliferation:** Low-cost, high-speed anti-ship missiles have spread to non-state actors and proxy forces. Incidents involving sea-skimming missiles like the C-802 or variants in restricted waters like the Red Sea prove that even minor actors can threaten unstealthy surface combatants.

(b) **Unmanned Systems and Drone Swarms:** The widespread use of low-cost loitering munitions and explosive Unmanned Surface Vessels (USVs) has altered littoral warfare. A modern warship must be able to detect, track, and destroy multiple small, fast-moving surface and aerial targets simultaneously.

(c) **Transnational Maritime Crime:** The IOR continues to deal with traditional security issues, including piracy originating from the Somali basin, maritime narcotics trafficking, weapons smuggling used to fund regional insurgencies, and widespread Illegal, Unreported, and Unregulated (IUU) fishing that depletes local marine resources.

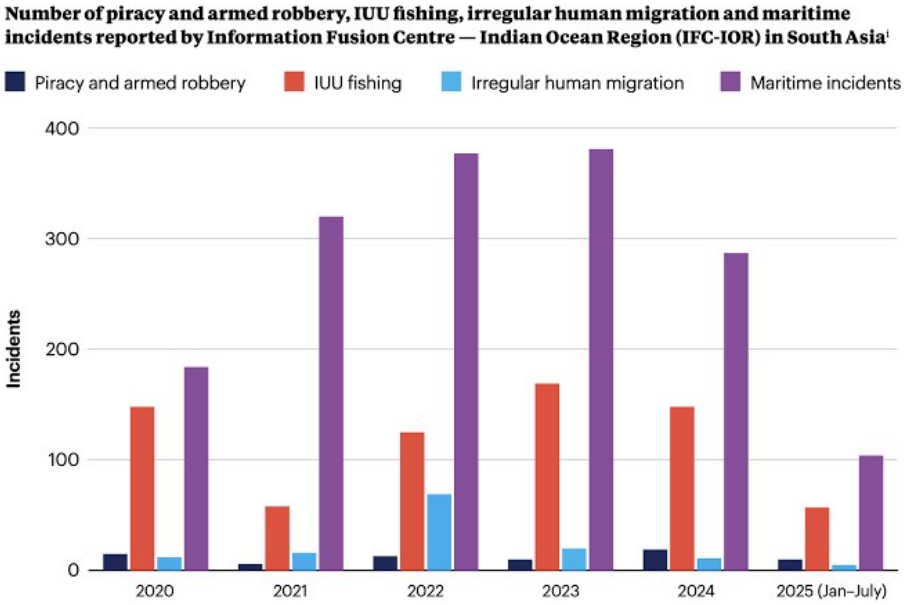


Figure 1.7 Indian Ocean maritime security and crime incident graph

1.4. TECHNICAL AND TACTICAL ROLE

In naval engineering terminology, a Corvette is defined as a surface combatant falling between an Offshore Patrol Vessel (OPV) and a Frigate. It typically features a displacement ranging from 1,000 to 3,500 tonnes, an overall length between 80 to 120 meters, and a hull optimized for speed, maneuverability, and shallow-draft operations.

The engineering and operational necessity of the corvette class stems from basic naval economics and hydrodynamic principles:

1.4.1. Littoral Hydrodynamics and Tactical Maneuverability

Large surface vessels like destroyers require significant operational drafts (often exceeding 6.5 meters), making them poorly suited for shallow coastal archipelagos, sandbanks, and restricted littoral channels. A corvette's shallow draft (typically under 4.5 meters) allows it to navigate these confined spaces easily. This shallow draft provides access to smaller ports and offers tactical cover near irregular coastlines, where land clutter can disrupt enemy radar tracking.

1.4.2. The Weapon-to-Displacement Efficiency Ratio

Modern structural design, lightweight composite superstructures, and compact weapon electronics allow engineers to mount heavy armament onto smaller hulls. A modern 3,500-tonne corvette can carry supersonic anti-ship cruise missiles, active towed array sonars, torpedo launchers, and vertical launch air defense networks—capabilities previously restricted to 6,000-tonne frigates.

1.4.3. Operational Cost Optimization

Building and maintaining a destroyer requires significant capital expenditure and large crew complements (often over 250 personnel). In contrast, a corvette integrates extensive automation—such as automated machinery control systems (AMCS) and smart damage control systems—which reduces crew requirements to fewer than 150 personnel. This lower manning requirement, combined with a smaller displacement, reduces fuel consumption and lowers overall lifecycle costs while keeping a highly capable combat asset at sea.

1.5. EVOLUTION OF FLEET HULLS

The development of the Indian Navy's corvette fleet marks a clear transition from basic foreign imports to advanced, domestically designed stealth platforms.

1.5.1. Legacy Strike Hulls

The older framework relied on the *Veer*-class missile boats (derived from the Soviet Project 1241RE Tarantul class) and the *Kora*-class guided-missile corvettes. These platforms offered high speeds and heavy forward fire networks, but lacked comprehensive multi-mission capabilities, offering limited air defense and minimal anti-submarine protection.

1.5.2. Project 28 (Kamorta Class)

To address the growing submarine threat in the region, the Directorate of Naval Design developed the **Project 28 *Kamorta*-class** ASW corvette. Built by Garden Reach Shipbuilders & Engineers (GRSE) in Kolkata, these vessels have a full-load displacement of 3,300 to 3,500 tonnes, an overall length of 109.1 meters, and a beam of 13.7 meters.

The Project 28 class introduced several key structural innovations:

(a) **Advanced Stealth Geometries:** The hull utilizes an 'X-form' cross-section with inclined superstructure bulkheads to scatter incoming radar waves, significantly lowering the vessel's radar cross-section.

(b) **Composite Superstructure Integration:** The upper superstructure incorporates Carbon Fiber Reinforced Plastic (CFRP) structural panels co-bonded to the steel hull structure. This lowered the ship's vertical center of gravity, which improved intact stability margins and reduced overall structural weight.



Figure 1.8 INS Kamorta antisubmarine corvette at sea

(c) **Acoustic Quieting and Vibration Isolation:** The main machinery is mounted on specialized raft isolation systems. This arrangement absorbs structural vibrations before they reach the hull plating, reducing the underwater radiated noise signature to hamper detection by enemy passive sonars.

(d) **Propulsion System:** The propulsion plant uses a Combined Diesel and Diesel (CODAD) configuration, pairing four Pielstick 12 PA6 STC diesel engines to drive twin controllable-pitch propellers, achieving a top speed of 32 knots and an economical cruising range of 4,000 nautical miles at 14 knots.\

1.5.3. Next-Generation Missile Vessels (NGMV)

Designed to replace the aging *Veer*-class, the NGMV is a specialized anti-surface strike vessel. Displacing roughly 1,437 tonnes with an overall length of 89 meters and a narrow beam of 12 meters, these ships are engineered for high-speed dashes.

The propulsion layout uses a Combined Diesel and Gas (CODAG) system that pairs a General Electric LM2500 gas turbine with two cruising diesel engines to power three Kongsberg Kamewa waterjets. This setup enables a maximum speed of 35 knots. The primary armament consists of eight BrahMos supersonic anti-ship cruise missiles carried in slant-launch configurations, combined with a 24-cell Vertical Launch Short Range Surface-to-Air Missile (VL-SRSAM) system for point defense.

1.5.4. Next Generation Corvettes (NGC)

The future core of the littoral combat fleet is the Next Generation Corvette program. Approved by the Ministry of Defence, this program involves constructing eight multi-role surface combatants, with GRSE serving as the lead shipyard for five hulls and Goa Shipyard Limited (GSL) building the remaining three.

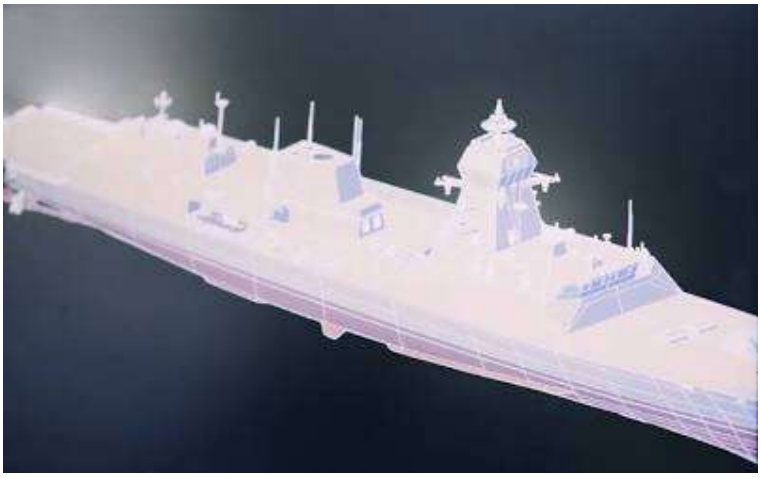


Figure 1.9 Indian Navy Next Generation Corvette (NGC) schematic concept

The NGC features a planned displacement of 3,800 to 4,000 tonnes, a length under 120 meters, and a beam under 15 meters. The propulsion system will utilize a robust CODAG network, combining two Rolls-Royce MT30 gas turbines with four indigenous 6 MW diesel engines to ensure a sustained cruising range of 4,000 nautical miles at 14 knots and top speeds exceeding 27 knots. The armament suite includes eight BrahMos cruise missiles, 24 vertical launch anti-missile defense cells, and dual triple-tube 324 mm lightweight torpedo launchers.

1.6. OPERATIONAL HISTORY AND CASE STUDIES

Small, heavily armed surface vessels have played a critical role in major operations conducted by the Indian Navy, validating their operational concept through direct combat success.

1.6.1. Operations Trident and Python (1971)

During the 1971 Indo-Pakistani War, the Indian Navy deployed Osa-class missile boats—the technological precursors to the modern missile corvette—in offensive operations. On the night of December 4, 1971, under Operation Trident, these agile

vessels executed a high-speed strike against the Pakistani naval hub at Karachi.



Figure 1.10 Karachi harbor burning during Operation Trident

Firing Soviet-supplied P-15 Termit anti-ship missiles, they sank the destroyer PNS *Khaibar* and the minesweeper PNS *Muhafiz*, while severely damaging merchant assets and shore-based fuel storage tanks. Days later, Operation Python struck the port again, crippling the local maritime logistics network. These actions established sea control for India in the northern Arabian Sea and demonstrated the striking power of small, fast missile combatants.

1.6.2. Operation Vijay and Operation Talwar (1999)

During the 1999 Kargil Conflict, the Indian Navy launched Operation Talwar to put aggressive pressure on Pakistan's maritime trade routes. Missile corvettes were deployed alongside larger destroyers to form a tight naval blockade in the northern Arabian Sea.

This forward deployment cut off commercial fuel routes to Karachi and discouraged potential naval escalation by the Pakistan Navy. The ability of corvettes to operate effectively near shallow coastal borders allowed larger capital ships to maintain broader sea-denial screens further out at sea.

1.6.3. Anti-Piracy and Escort Deployments (2000s – Present)

In response to the rise of piracy in the western Indian Ocean and the Gulf of Aden, the Indian Navy has maintained a continuous presence in these high-risk areas. Corvettes have proven highly effective for these missions. Their operational endurance allows them to conduct extended patrols, while their onboard helicopter facilities and rigid-hull inflatable boats (RHIBs) enable rapid boarding actions. Deploying corvettes for these tasks keeps international sea lanes open while preserving larger destroyers for primary defense duties elsewhere.

1.7. MODULAR DESIGN PRINCIPLES

A foundational aspect of modern corvette design is the adoption of **modular payload architectures**. Historically, warships were designed around fixed systems; weapons and sensors were permanently integrated into the structural fabric of the hull. This traditional approach meant that mid-life refits took years to complete, requiring extensive structural cutting, welding, and balance recalibrations.

Modern corvette engineering solves this issue using standardized interfaces—such as standardized containerized mission slots—which allow systems to be swapped out quickly:

1.7.1. Structural and Utility Standardization

Mission bays are engineered with uniform spatial dimensions, structural load foundations, standard electrical connections, and uniform cooling water systems. This design allows the vessel's primary role to be reconfigured within 48 hours at a forward naval base.

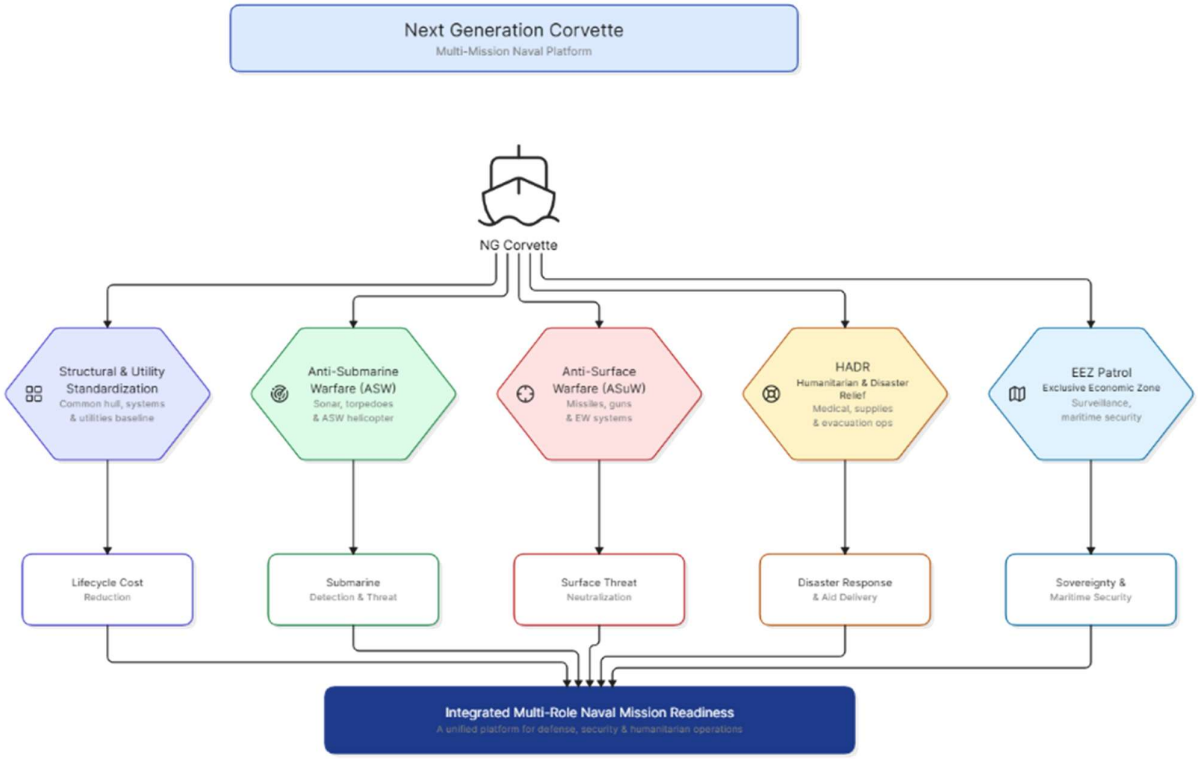


Figure 1.11 Concept of NGC

1.7.2. Anti-Submarine Warfare (ASW) Mode

The modular bay is fitted with an active towed array sonar system, processing racks, and lightweight torpedo storage systems.

1.7.3. Anti-Surface Warfare (ASuW) Mode

The ASW systems are removed and replaced with containerized surface-to-surface missile launch systems or long-range loitering munitions to boost surface striking power.

1.7.4. Humanitarian Assistance and Disaster Relief (HADR) Mode

The weapons modules are replaced with mobile medical stations, water purification machinery, and emergency supplies, allowing the corvette to act as a first responder during regional environmental crises.

Figure 1.12 next generation corvette

1.8. SIGNATURE MANAGEMENT SYSTEMS

Surviving in modern contested littorals requires a comprehensive approach to signature management. Because corvettes operate closer to hostile coastlines than larger frigates, they face higher exposure to land-based radar installations, thermal imaging networks, and shallow-water acoustic sensor arrays.

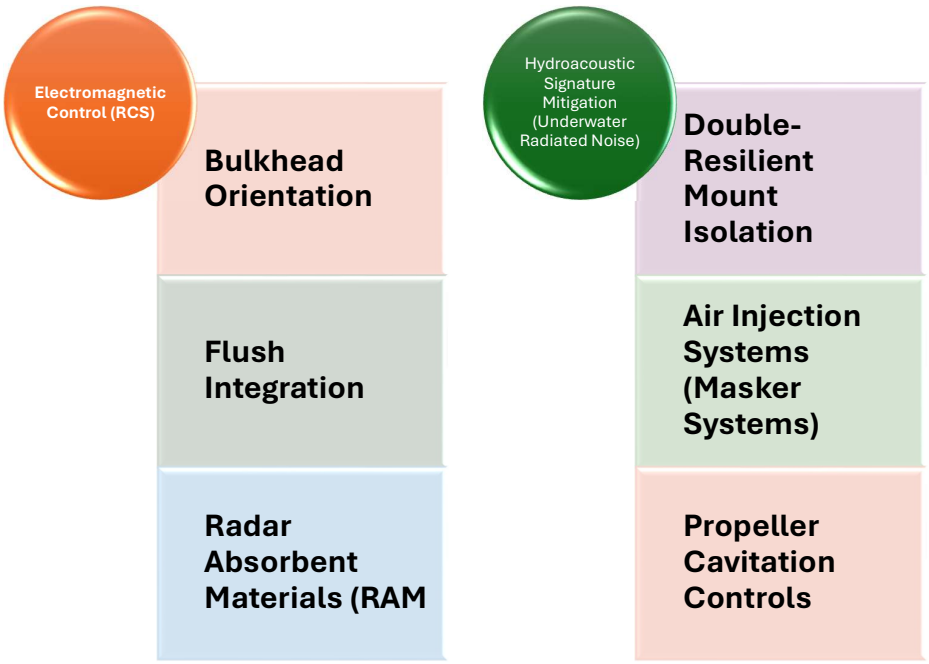


Figure 1.13 Signature Management System

1.8.1. Electromagnetic Control (RCS)

Reducing the vessel's Radar Cross-Section relies on geometric configuration and advanced material placement:

(a) **Bulkhead Orientation:** Superstructure slopes are maintained at angle configurations between 10 degrees to 15 degrees relative to the vertical plane. This angle prevents orthogonal radar reflections, scattering incoming electromagnetic waves away from the enemy receiver antenna.

(b) **Flush Integration:** Deck equipment—including mooring winches, bitts, and chocks—is positioned beneath the main deck or hidden behind clean bulwark screens.

(c) **Radar Absorbent Materials (RAM):** The superstructure is treated with specialized multi-layer dielectric and magnetic RAM coatings. These coatings convert incoming electromagnetic energy into thermal energy, significantly reducing the ship's overall radar detection range.

1.8.2. Hydroacoustic Signature Mitigation (Underwater Radiated Noise)

Passive acoustic sonars on enemy submarines represent a serious threat to surface hulls. Minimizing Underwater Radiated Noise requires isolated machinery layouts:

(a) **Double-Resilient Mount Isolation:** Main propulsion diesels and diesel generator sets are attached to an intermediate structural raft plate via elastomeric isolators. This assembly is then mounted to the ship's primary framing network using a second set of high-deflection rubber isolators, blocking structural vibrations from reaching the outer hull plating.

(b) **Air Injection Systems (Masker Systems):** Perforated air emitter bands are installed along the hull's exterior near the engine spaces. These bands discharge a continuous screen of micro-bubbles into the boundary layer, altering the local acoustic impedance and dampening internal machinery noise.

(c) **Propeller Cavitation Controls**: Propeller blade geometries are optimized using advanced computational fluid dynamics (CFD) to delay the onset of tip cavitation. Additionally, air is injected through the propeller hub and out along the blade edges (Pratt-Prairie systems) to collapse cavitation bubbles smoothly, reducing high-frequency acoustic emissions.

1.9. THE STRATEGIC IMPERATIVE FOR INDIGENOUS SHIPBUILDING (*AATMANIRBHAR BHARAT*)

Transitioning from foreign procurement to absolute domestic self-reliance (Aatmanirbhar Bharat) is a key requirement for modern Indian warship design. Historically, relying on foreign shipyards created significant vulnerabilities:

1.9.1. Geopolitical Sanction Risks

Relying on foreign suppliers leaves naval construction programs vulnerable to sudden changes in international relations, technology transfer restrictions, and unilateral sanctions. These external

disruptions can halt spare parts pipelines during periods of active conflict.

1.9.2. Lifecycle and Configuration Control

When a warship is designed and built domestically, the navy retains absolute control over its configuration management data. This allows engineers to modify structural frames, update software code in the Combat Management System, and replace aging electronic subsystems without requiring permission or documentation from a foreign original equipment manufacturer (OEM).

1.9.3. Domestic Industrial Development

Modern naval construction serves as an economic multiplier. The Project 28 class, for example, achieved an indigenous content level close to 90%. This shift supported a domestic defense supply network by sourcing equipment locally, including domestic warship steel (DMR 249A), indigenous hull-mounted sonars (HUMSA), domestic 3D surveillance radars (Revathi), and indigenous chaff decoy systems (Kavach).

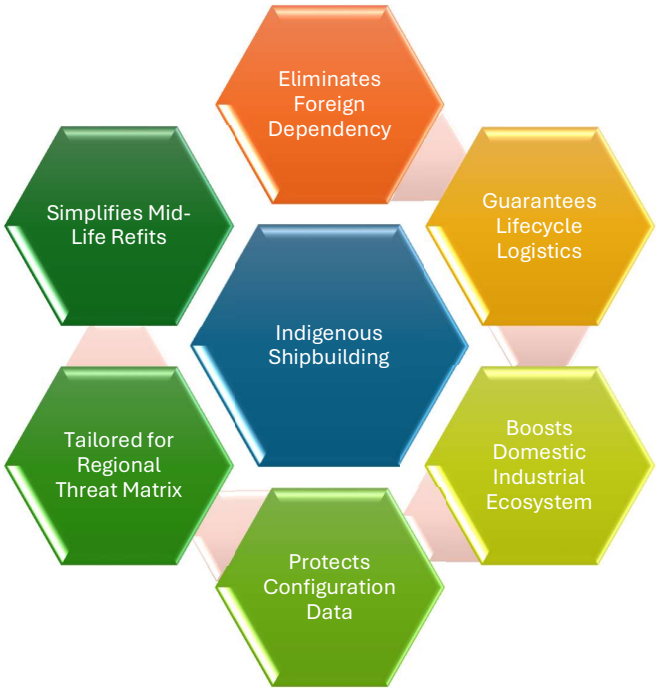


Figure 1.14 Advantages of Indigenous Ship Building

1.10. ORGANISATION OF THE THESIS

The project will progress along the lines of the chapters as follows:

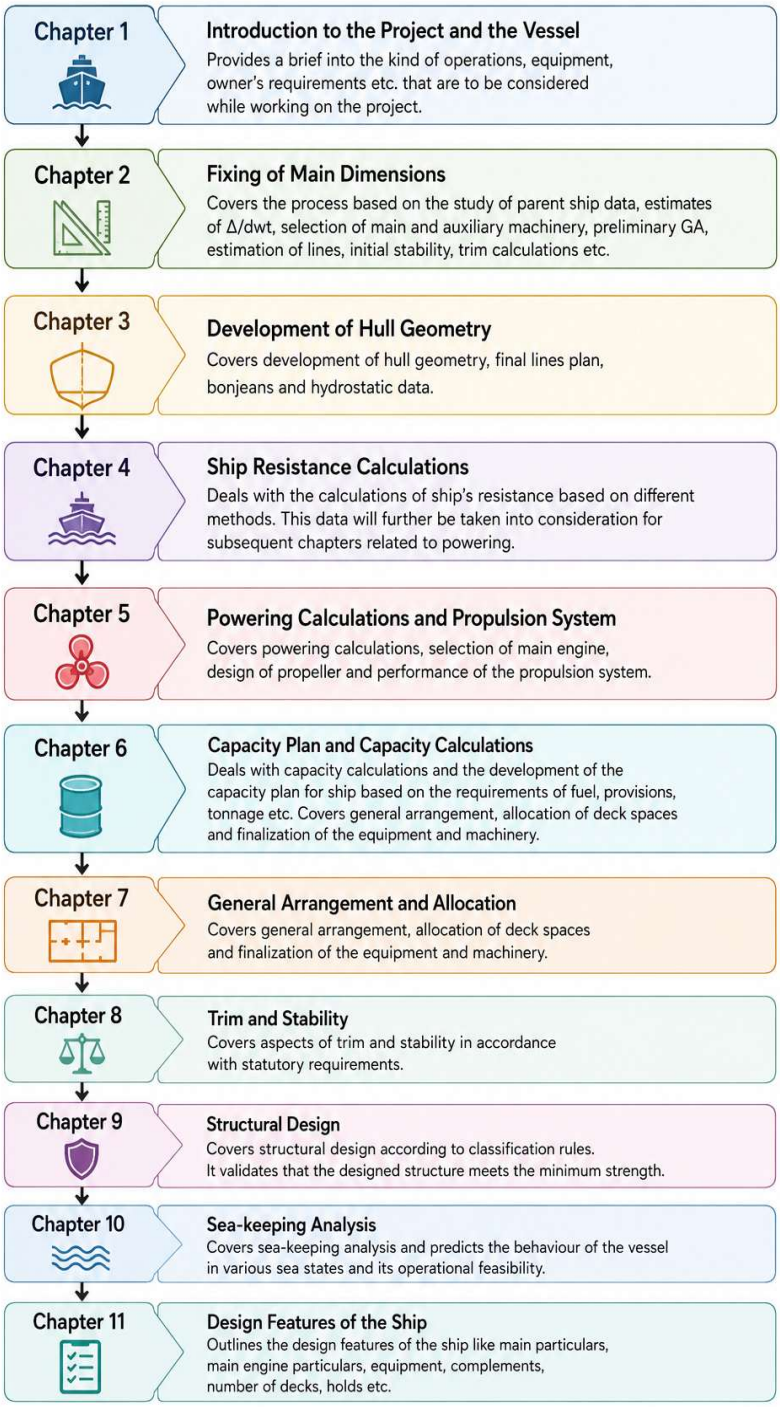


Figure 1.15 Organisation of the thesis

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1.12. OUTLINE SPECIFICATIONS

General Characteristics	
Type	Multi-Role Corvette
Role	Anti-Submarine Warfare, Anti-Surface Warfare, Air Defence, Maritime Surveillance, EEZ Patrol and Mine Warfare Support
Displacement	3,500 – 4,000 tonnes
Length overall	112 – 118 m
Beam	15 – 17 m
Draft	5.0 – 5.5 m
Speed	26–28 Knots (Maximum) 15 Knots (Cruising)
Range	4,000 – 5,000 nautical miles at 15 Knots
Endurance	20 – 30 days
Combat Capability	
Primary Mission	Multi-Role Naval Operations
Secondary Missions	Maritime Security, Coastal Defence, Convoy Escort, Search & Rescue and Humanitarian Assistance & Disaster Relief
Operational Capability	Anti-Submarine Warfare, Anti-Surface Warfare, Local Area Air Defence and Mine Warfare Support
Mission Flexibility	Modular payloads and future mission upgrades
Aviation Facilities	
Flight Deck	Capable of operating one medium naval helicopter
Hangar	Enclosed Hangar for one helicopter and UAV facilities
Mission Systems	
Mission Bay	Multi-purpose Mission Space for modular equipment
Unmanned Systems	Provision for UAV, USV and AUV Operations
Boat Facilities	RHIB Launch and Recovery Arrangement
Accommodation	
Crew	70 – 90 personnel
Mission Personnel	20 – 35 specialists
Medical Facilities	Sickbay, Isolation Room and First Aid Facility
Propulsion	
Type	CODAD
Propulsion	Twin-Shaft Configuration
Bow Thrusters	If required
Power Generation	Diesel Gen Sets
Self Defense & Survivability	

Weapons	Medium Calibre Naval Gun, Close-In Weapon System (CIWS), Surface-to-Air Missiles, Anti-Ship Missiles and Lightweight Torpedoes
Electronic Systems	ESM Suite, Decoy Launchers, Electronic Countermeasures and Electro-Optical Fire Control

Design Considerations	
Standards	Naval Classification Rules and Naval Ship Code
Stealth Design	Reduced Radar Cross Section, Infrared Signature Reduction and Low Acoustic Signature
Modular Design	Yes
Future Growth	Reserved Space, Weight and Power Margins for future weapons and sensors
Automation	High Platform Automation with Reduced Crew Requirements
Survivability	Enhanced Shock Resistance, Redundant Machinery Arrangement and Improved Damage Control
Sea Keeping	Optimized for sustained operations in littoral and open sea conditions up to Sea State 6